Cairo University
Faculty of Engineering

MTHG002
Midterm Exam

Fall 2023
Duration: 60 minutes

ظلل إجاباتك بوضوح في الدوائر المخصصة فقط. حافظ على نظافة ورقتك.

Shade your answers clearly in the bubble sheet only. Keep your paper clean.

1. A matrix which is obtained by preforming six elementary row operations on I_n is:
(a) nonsingular (b) elementary (c) singular (d) rectangular (e) None

2. A, B, and C are $n \times n$ matrices. A is diagonal, B is lower triangular, and C is upper triangular. $(C + A)B^T$ is:
(a) lower triangular (b) symmetric
(c) upper triangular (d) diagonal (e) None

3. If A is an $n \times m$ matrix and B is an $n \times n$ skew-symmetric matrix, then $A^T BA$ is:
(a) symmetric (b) skew-symmetric
(c) diagonal (d) triangular (e) None

4. If $x = -2$ is a root of the equation $2x^5 + 5x^4 + a_3x^3 - 23x^2 + 79x - 30 = 0$, then a_3 is equal to:
(a) -33 (b) 33 (c) 16.5 (d) -16.5 (e) None

5. If the remainder of dividing $P(x)$ by $(x^2 + 3x)$ is equal to $(3x + 5)$, then $P(-3)$ is equal to:
(a) -2 (b) -3 (c) -5 (d) -4 (e) None

6. If $A = \begin{bmatrix} 1 & 3 & -7 \\ -2 & 1 & r \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 1 and column 2 of AA^T is equal to:
(a) $7r - 1$ (b) $-7 - 2r$ (c) $-7r + 1$ (d) $7 + 2r$ (e) None

7. The linear system $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \underline{X} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ w \end{bmatrix}$ will have a unique solution if w is equal to:
(a) -8 (b) -2 (c) -4 (d) -6 (e) None

8. If $P(x) = x^3 + 4x^2 - 8x + a_0$ is divided by $(x - (2 - i))$, then the remainder is:
(a) $a_0 - 4 - 18i$ (b) $a_0 - 2 - 19i$ (c) $a_0 + 2 - 21i$ (d) $a_0 + 4 - 22i$ (e) None

9. If $(A^T + 3I)^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & y \end{bmatrix}$, $y \neq 0$, then the entry in row 2 and column 1 of A is equal to:

(a) $-4/y$ (b) $4/y$ (c) $-1/y$ (d) $1/y$ (e) None

10. The linear system $\begin{aligned} x - 4y + 2z &= 0 \\ x + 3y - 4z &= 0 \\ rx + sy + tz &= 0 \end{aligned}$ will have nontrivial solutions if:

(a) $2r + s + t = 0$ (b) $r - 2s + 3t = 0$

(c) $2r + 5s + 2t = 0$ (d) $10r + 6s + 7t = 0$ (e) None

$x = -1$ is a repeated root of multiplicity 2 of the equation $3x^5 + 2x^4 - 22x^3 - 32x^2 + a_1x + a_0 = 0$.

11. The value of a_1 is:

(a) 5 (b) -5 (c) 1.67 (d) -1.67 (e) None

12. The value of a_0 is:

(a) -6 (b) 2 (c) 6 (d) -2 (e) None

13. Consider the linear system $\begin{aligned} x + 4y + 3z &= 2 \\ x + 5y + z &= -4 \\ 3x + 2y + 29z &= 66 \end{aligned}$. If $z = t, t \in \mathbb{R}$, then x is equal to:

(a) $26 + 11t$ (b) $13 + 4t$ (c) $13 - 4t$ (d) $26 - 11t$ (e) None

14. The linear system $\begin{aligned} x - 2y + 4z &= a \\ 2x - 3y + 11z &= b \\ 3x - 3y + 21z &= c \end{aligned}$ will be consistent if:

(a) $c = 3a - 3b$ (b) $c = -5a + 4b$

(c) $c = 5a - 4b$ (d) $c = -3a + 3b$ (e) None

15. Consider the equation $x^3 + x^2 + ax + b = 0$, where a and b are nonzero real numbers. The equation will have one positive real root if:

(a) b is positive (b) a and b have the same sign

(c) b is negative (d) a and b have opposite signs (e) None

16. The entry in row 1 and column 7 of the reduced row echelon form (RREF) of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 2 & 5 & 0 & 9 & 2 & 4 & 11 \\ 3 & 11 & 1 & 15 & 2 & 11 & 14 \end{bmatrix}$ is equal to:

(a) -3 (b) -4 (c) -2 (d) -1 (e) None

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Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 11 \\ 4 & 5 & f \end{bmatrix}$ where $f \neq 32$

17. The entry in row 2 and column 3 of U is equal to:
(a) -11 (b) -5 (c) -6 (d) -4 (e) None

18. The entry in row 3 and column 3 of L is equal to:
(a) $1/(f-3)$ (b) $f-3$ (c) $f-4$ (d) $1/(f-4)$ (e) None

19. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 2 & -3 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:
(a) -8 (b) $-1/8$ (c) -7 (d) $-1/7$ (e) None

Consider the linear system
$$\begin{aligned} x + 2y + 2z &= 1 \\ 2x + 2y + 3z &= -1, k \neq 16. \\ 3x + 4y + kz &= 1 \end{aligned}$$

20. The value of y is equal to:
(a) $(3k-25)/(k-16)$ (b) $(4k-13)/(k-16)$
(c) $(3k-28)/(k-16)$ (d) $(4k-15)/(k-16)$ (e) None

21. The value of x is equal to:
(a) $(-5k+32)/(k-16)$ (b) $(-5k+37)/(k-16)$
(c) $(-3k+22)/(k-16)$ (d) $(-3k+25)/(k-16)$ (e) None

The matrix $A = \begin{bmatrix} 1 & 2 & 6 & 3 \\ 2 & 5 & 15 & 11 \\ 3 & 2 & 7 & -7 \\ -2 & -1 & -1 & k \end{bmatrix}$, where $k \neq 17$, is nonsingular.

22. The entry in row 1 and column 4 of A^{-1} is:
(a) $9/(k-17)$ (b) $7/(k-17)$
(c) $5/(k-17)$ (d) $3/(k-17)$ (e) None

23. The entry in row 1 and column 1 of A^{-1} is:
(a) $5(k+34)/(k-17)$ (b) $5(k+16)/(k-17)$
(c) $5(k+7)/(k-17)$ (d) $5(k+25)/(k-17)$ (e) None

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Algebra

1. (a) Non-Singular

2. (c) upper tri. $(C+A)B^T = CB^T + AB^T = 4+4 = 4$

3. (b) Skew-Symm. ∵ B is skew ∴ $B^T = -B$, $(A^TBA)^T = A^TB^TA = A^T(-B)A$

$= -A^TBA$ ∴ A^TBA is skew

4. (a) 12 $(-2)^5 + 5(-2)^4 + a_3(-2)^3 - 23(-2)^2 + 79(-2) - 30 = 0$ ∴ $a_3 = -33$ (a)

5. (-4) $\frac{P(x)}{x^2+3x} = Q + \frac{3x+5}{x^2+3x}$ $\xrightarrow{R}$ ∴ $P(-3) = 0 + 3x - 3 + 5 = -4$

6. (c) $-7r+1$ $AA^T = \begin{bmatrix} 1 & 3 & -7 \\ _ & _ & _ \end{bmatrix}\begin{bmatrix} -2 & \\ 1 & \\ r & \end{bmatrix} = \begin{bmatrix} -1-7r & \\ _ & _ \end{bmatrix}$

7. (a) -6
$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & w \end{array}\right] \xrightarrow{-r_2+r_4\to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -3 & w \end{array}\right] \xrightarrow{3r_3+r_4\to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & -10 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & w+6 \end{array}\right]$$ ∴ $w = -6$

8. (b) $a_0 - 2 - 19i$

$2-i$	1	4	-8	a_0
	1	$2-i$	$11-8i$	$-2-19i$
	1	$6-i$	$3-8i$	$a_0-2-19i$

9. $(A^T+3I) = \begin{bmatrix} 1 & 4 \\ 0 & 9 \end{bmatrix}^{-1}$ ∴ $A^T = \begin{bmatrix} 1 & -\frac{4}{9} \\ 0 & \frac{1}{9} \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -\frac{4}{9} \\ 0 & \frac{1}{9}-3 \end{bmatrix}$ ∴ $A = \begin{bmatrix} -2 & 0 \\ -\frac{4}{9} & \frac{1}{9}-3 \end{bmatrix}$

$-\frac{4}{9}$

10. (d) $10r+6s+7t=0$

$$\left(\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 3 & -4 & 5 & 0 \\ r & s & t & 0 \end{array}\right) \xrightarrow{-r_1+r_2\to r_2,\ (-r)r_1+r_3\to r_3} \left(\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 7 & -6 & 0 \\ 0 & s+4r & t-2r & 0 \end{array}\right)$$

$$\xrightarrow{\frac{1}{7}r_2\to r_2} \left(\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & -\frac{6}{7} & 0 \\ 0 & s+4r & t-2r & 0 \end{array}\right) \xrightarrow{-(s+4r)r_2+r_3\to r_3} \left(\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & -\frac{6}{7} & 0 \\ 0 & 0 & \frac{1}{7}(s+4r)+t-2r & 0 \end{array}\right)$$

∴ $\frac{6}{7}(s+4r) + t - 2r = 0$

∴ $6s + 24r + 7t - 14r = 0$

∴ $10r + 6s + 7t = 0$

11. $a_1 = -5$ (b), $a_0 = 6$ (c)

12.

	3	2	-22	-32	a_1	a_0
-1		-3	1	21	11	$-a_1-11$
	3	-1	-21	-11	a_1+11	a_0-a_1-11
		-3	4	17	-6	$-a_1-5$
-1	3	-4	-17	6	a_1+5	a_0-2a_1-6

$a_0 - a_1 - 11 = 0$, $a_0 - 2a_1 - 16 = 0$ $\therefore a_0 = 6$, $a_1 = -5$

13. (d)
$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 1 & 5 & 1 & -4 \\ 3 & 2 & 29 & 66 \end{array}\right) \xrightarrow[-3r_1+r_3\to r_3]{-r_1+r_2\to r_1} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & -6 \\ 0 & -10 & 20 & 60 \end{array}\right) \xrightarrow{10r_2+r_3\to r_3} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$z = t$ $\therefore y - 2t = -6$ $\therefore y = 2t - 6$, $x + 4(2t-6) + 3t = 2$ $\therefore x = 26 - 11t$ (d)

14. (d) $-3a + 3b$
$$\left(\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 2 & -3 & 11 & b \\ 3 & -3 & 21 & c \end{array}\right) \xrightarrow[-3r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \left(\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 3 & 9 & c-3a \end{array}\right)$$

$3(b-2a) = c - 3a$ $\therefore$ $c = 3b - 3a$ (d)

15. (d) a and b have opposite signs (so there'll be one change in signs).

16. (a) -3
$$\left(\begin{array}{ccccccc} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 2 & 5 & 0 & 9 & 2 & 4 & 8 \\ 3 & 8 & 1 & 15 & 2 & 8 & 14 \end{array}\right) \xrightarrow[-3r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \left(\begin{array}{ccccccc} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 & 0 & 2 & 2 \\ 0 & 2 & 4 & 3 & -1 & 5 & 5 \end{array}\right)$$

$$\xrightarrow[-2r_2+r_3\to r_3]{-2r_2+r_1\to r_1} \left(\begin{array}{ccccccc} 1 & 0 & -5 & 2 & 1 & -3 & -1 \\ 0 & 1 & 2 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right) \xrightarrow[-2r_3+r_1\to r_1]{-r_3+r_2\to r_2} \left(\begin{array}{ccccccc} 1 & 0 & -5 & 0 & 3 & -5 & -3 \\ 0 & 1 & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right)$$

17. (b) -5

18. (b) F-3
$$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 1 & F-3 \end{array}\right], \quad \left[\begin{array}{ccc} 2 & 2 & 4 \\ 3 & 2 & 11 \\ 4 & 5 & F \end{array}\right] \xrightarrow{\frac{1}{2}r_1\to r_1} \left[\begin{array}{ccc} 1 & 1 & 2 \\ 3 & 2 & 11 \\ 4 & 5 & F \end{array}\right] \xrightarrow[-4r_1+r_3\to r_3]{-3r_1+r_2\to r_2} \left[\begin{array}{ccc} 1 & 1 & 2 \\ 0 & -1 & 5 \\ 0 & 1 & F-8 \end{array}\right]$$

$$\xrightarrow{-r_2\to r_2} \left[\begin{array}{ccc} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 1 & F-8 \end{array}\right] \xrightarrow{-r_2+r_3\to r_3} \left[\begin{array}{ccc} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 0 & F-3 \end{array}\right] \xrightarrow{r_3\times\frac{1}{F-3}\to r_3} \left[\begin{array}{ccc} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{array}\right]$$

19. (a) -8

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 1 & 2 & 1 & -8 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix} \xrightarrow[-4r_1+r_3\to r_3]{\frac{r_1}{2}\to r_1,\ -2r_1+r_2\to r_2} \begin{bmatrix} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix}$$

$$\xrightarrow[-2r_2+r_4\to r_4]{-r_2+r_3\to r_3} \begin{bmatrix} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & +1 \\ 0 & 0 & 1 & -7 \end{bmatrix} \xrightarrow{-r_3+r_4\to r_4} \begin{bmatrix} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -8 \end{bmatrix} \xrightarrow{\frac{1}{-8}r_4\to r_4} 4$$

20. (c) $\dfrac{3k-28}{k-16}$

21. (a) $\dfrac{-5k+32}{k-16}$

$$\left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 3 & 9 & -1 \\ 3 & 4 & k & 1 \end{array}\right) \xrightarrow[-3r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -1 & 5 & -3 \\ 0 & -2 & k-6 & -2 \end{array}\right) \xrightarrow{-r_2\to r_2} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & -5 & 3 \\ 0 & -2 & k-6 & -2 \end{array}\right)$$

$$\xrightarrow[2r_2+r_3\to r_3]{-2r_2+r_1\to r_1} \left(\begin{array}{ccc|c} 1 & 0 & 12 & -5 \\ 0 & 1 & -5 & 3 \\ 0 & 0 & k-16 & 4 \end{array}\right) \xrightarrow{\frac{1}{k-16}r_3\to r_3} \left(\begin{array}{ccc|c} 1 & 0 & 12 & -5 \\ 0 & 1 & -5 & 3 \\ 0 & 0 & 1 & \frac{4}{k-16} \end{array}\right) \xrightarrow[-12r_3+r_1\to r_1]{5r_3+r_2\to r_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -5-\frac{48}{k-16} \\ 0 & 1 & 0 & 3+\frac{20}{k-16} \\ 0 & 0 & 1 & \frac{4}{k-16} \end{array}\right)$$

$y = 3 + \dfrac{20}{k-16} = \boxed{\dfrac{3k-28}{k-16}}$ (c), $x = -5 - \dfrac{48}{k-16} = \boxed{\dfrac{-5k+32}{k-16}}$ (a)

22. (b) 23. (d)

$$\left(\begin{array}{cccc} 1 & 2 & 6 & 3 \\ 2 & 5 & 15 & 11 \\ 3 & 2 & 7 & -7 \\ -2 & -1 & -1 & k \end{array}\right) \left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} -2r_1+r_2\to r_2 \\ -3r_1+r_3\to r_3 \\ 2r_1+r_4\to r_4 \end{array}$$

$$\left(\begin{array}{cccc} 1 & 2 & 6 & 3 \\ 0 & 1 & 3 & 5 \\ 0 & -4 & -11 & -16 \\ 0 & 3 & 11 & 6+k \end{array}\right) \left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} -2r_2+r_1\to r_1 \\ 4r_2+r_3\to r_3 \\ -3r_2+r_4\to r_4 \end{array}$$

$$\left(\begin{array}{cccc} 1 & 0 & 0 & 7 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 2 & k-9 \end{array}\right) \left(\begin{array}{cccc} 5 & -2 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -11 & 4 & 1 & 0 \\ 8 & -3 & 0 & 1 \end{array}\right) \begin{array}{l} -3r_3+r_2\to r_2 \\ -2r_3+r_4\to r_4 \end{array}$$

$$\left(\begin{array}{cccc} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & k-17 \end{array}\right) \left(\begin{array}{cccc} 5 & -2 & 0 & 0 \\ 31 & -11 & -3 & 0 \\ -11 & 4 & 1 & 0 \\ 30 & -11 & -2 & 1 \end{array}\right) \frac{1}{k-17}r_4\to r_4$$

$$\left(\begin{array}{cccc} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right) \left(\begin{array}{cccc} 5 & -2 & 0 & 0 \\ 31 & -11 & -3 & 0 \\ -11 & 4 & 1 & 0 \\ \frac{30}{k-17} & \frac{-11}{k-17} & \frac{-2}{k-17} & \frac{1}{k-17} \end{array}\right) \begin{array}{l} -4r_4+r_3\to r_3 \\ 7r_4+r_2\to r_2 \\ 7r_4+r_1\to r_1 \end{array}$$

$$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \left(\begin{array}{cccc} \frac{210}{k-17}+5 & \frac{-77}{k-17} & -2 & \frac{-14}{k-17} \\ & & & \frac{7}{k-17} \end{array}\right)$$

$$\frac{210}{k-17} + 5 = \frac{5k+125}{k-17} = \frac{5(k+25)}{k-17}$$

Eng/Youssef Hesham